

Yashasvi Karnena

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Germany

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Professional Experience

- Apr 23 – Present **fka GmbH** – Aachen, Germany
Development Engineer (Software)
- Developed ISO-26262-qualified real-time control software components using TargetLink and MATLAB/Simulink.
 - Built Functional Mockup Units (FMUs) for validating ECU firmware against controlled plant models.
 - Investigated Nonlinear control methods for PMSM position control in Steer-By-Wire (SbW) systems.
 - Conducted HiL, MiL, and SiL testing of functional software.
 - Researched and implemented concepts for cloud based plant parameter estimation and adaptive controller tuning of Software Defined Vehicles (SDVs).
- Apr 22 – Mar 23 **fka GmbH** – Aachen, Germany
Master Thesis Student
- Implemented position trajectory tracking controller using nonlinear 2DOF control.
 - Developed nonlinear MPC trajectory planner with State-Lattice planners for solver warm-starting to tackle non-convexities.
 - Proposed a state-machine based sequential motion cooperation algorithm for Connected Autonomous Vehicles.
- Nov 21 – Mar 22 **fka GmbH** – Aachen, Germany
Intern
- Designed cascaded position control loops and extended Kalman filters (Parameter Estimation) for Steer-by-Wire systems.
 - Developed C/C++ estimators and filters; deployed on PXROS RTOS.
 - Debugged Hardware, Firmware and control loop problems.
- May 21 – Oct 21 **Institute for Electrical Machines, RWTH Aachen** – Aachen, Germany
Student Assistant
- Created Simulink exercises for lab course – *Dynamics of Electrical Machines*.
 - Implemented Simulink based bearing failure analysis models arising out of inverter harmonics.
- May 18 – Apr 19 **Altigreen Propulsion Labs Private Limited** – Bengaluru, India
Junior Electrical Engineer
- Developed firmware for telematics units and Inverter FOC algorithms in Simulink.
 - Implemented toolchains using CMake/Jenkins for TI C2000 and ARM Cortex-M microcontroller platforms.
- Apr 17 – Sep 17 **Defense Avionics Research Establishment** – Bengaluru, India
Intern
- Designed power distribution boards using LTSpice and Altium.
 - Implemented avionics logic on FPGAs; developed analog/digital filters.

Education

Oct 19 – May 23	RWTH Aachen University – Aachen, Germany <i>M.Sc in Electrical Engineering, IT & Computer Engineering</i> Major: Systems and Automation (Total Grade – 2,2)
Jul 14 – May 18	Vellore Institute of Technology – Vellore, India <i>B.Tech in Electrical and Electronics Engineering</i> (Total Grade – 1,9)

Skills

Programming:	C/C++, Java, Python, Golang, Bash, LaTeX, CMake
Code Generation:	MATLAB/Simulink Coder, dSPACE TargetLink, MathWorks Target Language Compiler
Simulation:	MATLAB, Simulink, LabView, PLECS, LTSpice, Simscape Language
EDA Tools:	KiCAD, Altium Designer, Eagle PCB
Testing:	CANoe, CANape, ControlDesk, PCAN-View
IT Tools:	Git, Jira, MS Office, IBM DOORS, PTC Windchill, Citavi
Languages:	English (C2), German (B1), Telugu (C2), Hindi (C1)

Professional Training

Oct 24	Automotive Cyber Security Training – NobleProg GmbH
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Extracurricular Activities

Apr 22 – Sep 24	Tachyon Hyperloop – RWTH Aachen University ○ Implemented Telemetry, Brakes Control and, Thermal Control hardware using Altium Designer. ○ Developed high voltage power electronics package for 3-Phase propulsion control. ○ Developed firmware using MATLAB/Simulink for TI C2000 microcontrollers. ○ Managed finances and projects as a board member and mentored new recruits in electronics design.
Feb 15 – Sep 17	Pravega Racing – Vellore Institute of Technology ○ Developed Electronic Throttle controller. ○ Tuned Engine Control Units and built race specification wiring harness. ○ Developed Traction Control Unit for race car launch control.

Awards

Aug 24	European Hyperloop Week – Best Electronics Package
Jan 17	Formula Student India – Best Engineering Design
May 13	NASA Space Settlement Contest – Honorable Mention

Yashasvi Karnena
Aachen, September 15, 2025